

Sequences and Series of Functions and Algebras

Write Up

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1 Definitions

Definition 1. We say that a sequence of functions $\{f_n\}, n = 1, 2, 3, \dots$ converges uniformly on E to a function f if for every $\epsilon > 0$, there is an integer N such that $n \geq N$ implies

$$|f_n(x) - f(x)| < \epsilon \quad (1)$$

for all $x \in E$.

Definition 2. Let $\{f_n\}$ be a sequence of functions defined on a set E . We say that $\{f_n\}$ is pointwise bounded on E if the sequence $\{f_n(x)\}$ is bounded for every $x \in E$, that is, if there exists a finite-valued function ϕ defined on E such that

$$|f_n(x)| < \phi(x) \quad (x \in E, n = 1, 2, 3, \dots). \quad (2)$$

We say that $\{f_n(x)\}$ is uniformly bounded on E if there exists a number M such that

$$|f_n(x)| < M \quad (x \in E, n = 1, 2, 3, \dots). \quad (3)$$

Definition 3. A family $\mathcal{F}$ of complex functions f defined on a set E in a metric space (X, d) is said to be equicontinuous on E if for every $\epsilon > 0$ there exists a $\delta > 0$ such that

$$|f(x) - f(y)| < \epsilon \quad (4)$$

whenever $d(x, y) < \delta$, $x \in E$, $y \in E$, and $f \in \mathcal{F}$.

Definition 4. A family $\mathcal{A}$ of complex functions defined on a set E is said to be an algebra if for all $f \in \mathcal{A}$ and $g \in \mathcal{A}$

- (i) $f + g \in \mathcal{A}$
- (ii) $fg \in \mathcal{A}$
- (iii) $cf \in \mathcal{A}$ for any complex constant c .

Definition 5. Suppose $\mathcal{A}$ is an algebra. $\mathcal{A}$ is said to separate points if for any pair of distinct points $x_1, x_2 \in E$ there exists a function $f \in \mathcal{A}$ such that $f(x_1) \neq f(x_2)$.

Definition 6. Suppose $\mathcal{A}$ is an algebra. $\mathcal{A}$ vanishes at no point in E if for any $x_1 \in E$ there exists a function $f \in \mathcal{A}$ such that $f(x_1) \neq 0$.

2 Worked Examples

Example 1. The set $\mathcal{A} = \{0\}$ is an algebra (we will denote it as the trivial algebra) that does not separate points and trivially does not have the property that $\mathcal{A}$ vanishes at no point in E .

Example 2. The set $\mathcal{A} = \{f(x) = c | c \in \mathbb{C}\}$ of all constant functions is an algebra that has the property that $\mathcal{A}$ vanishes at no point in E (for any point $x \in E$ the function $f(x) = 1$ will suffice) but does not separate points as every function $f \in \mathcal{A}$ satisfies $f(x) = f(y)$ for all points $x \in E$.

3 Theorems

Theorem 1. Every uniformly convergent sequence of bounded functions is uniformly bounded.

Proof. Suppose that $\{f_n\}$ is uniformly convergent and converges to f and each f_n is bounded. Thus for each f_n there exists M_n such that $|f_n(x)| \leq M_n$ for all $x \in E$. First we will show that $f(x)$ is bounded. Since $\{f_n\}$ is uniformly convergent there exists an N such that $n \geq N \implies |f_n(x) - f(x)| < 1$. Thus $|f(x)| = |f(x) - f_n(x) + f_n(x)| \leq |f(x) - f_n(x)| + |f_n(x)| \leq 1 + M_n$ so $f(x)$ is bounded and so $\sup |f(x)|$ exists. Now consider $\max\{M_1, \dots, M_n, \sup |f(x)| + 1\}$. If $m \leq N$ then $|f_m(x)| \leq M_m \leq \max\{M_1, \dots, M_n, \sup |f(x)| + 1\}$ and if $m > N$ then $|f_m(x)| \leq \sup |f(x)| + 1 \leq \max\{M_1, \dots, M_n, \sup |f(x)| + 1\}$. Thus $\max\{M_1, \dots, M_n, \sup |f(x)| + 1\}$ is an upper bound for $|f_n(x)|$ for all n and $x \in E$. Since this bound does not depend on x , we have that $\{f_n\}$ is uniformly bounded. $\square$

Theorem 2. If $\{f_n\}$ and $\{g_n\}$ converge uniformly on a set E then $\{f_n + g_n\}$ converges uniformly on E . In addition if $\{f_n\}$ and $\{g_n\}$ are sequences of bounded functions then $\{f_n g_n\}$ converges uniformly on E .

Proof. Suppose that $\{f_n\}$ and $\{g_n\}$ converge uniformly on a set E . Then $\forall \epsilon > 0, \exists N_1$ s.t. $n \geq N_1 \implies |f_n(x) - f(x)| < \frac{\epsilon}{2}$ for all $x \in E$ and $\forall \epsilon > 0, \exists N_2$ s.t. $n \geq N_2 \implies |g_n(x) - g(x)| < \frac{\epsilon}{2}$ for all $x \in E$. Choose $N = \max\{N_1, N_2\}$. Then $n \geq N \implies |f_n + g_n - (f + g)| = |f_n - f + g_n - g| \leq |f_n - f| + |g_n - g| < \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$. Since this N does not depend on x we have that $\{f_n + g_n\}$ converges uniformly on E . Now suppose additionally that $\{f_n\}$ and $\{g_n\}$ are sequences of bounded functions. Then $\exists M_1$ such that $|f_n(x)| \leq M_1$ and $\exists M_2$ such that $|g_n| \leq M_2$ (since g is bounded per theorem 1). We also have that $\forall \epsilon > 0, \exists N_1$ s.t. $n \geq N_1 \implies |f_n(x) - f(x)| < \frac{\epsilon}{2M_2}$ and $\forall \epsilon > 0, \exists N_2$ s.t. $n \geq N_2 \implies |g_n(x) - g(x)| < \frac{\epsilon}{2M_1}$. Choose $N = \max\{N_1, N_2\}$. Then $n \geq N \implies |f_n g_n - f g| = |f_n g_n - f_n g + f_n g - f g| = |f_n(g_n - g) + g(f_n - f)| \leq |f_n| |g_n - g| + |g| |f_n - f| \leq M_1 * \frac{\epsilon}{2M_1} + M_2 * \frac{\epsilon}{2M_2} = \frac{\epsilon}{2} + \frac{\epsilon}{2} = \epsilon$. Since N does not depend on x it follows that $\{f_n g_n\}$ converges uniformly on E . $\square$

Theorem 3. For $n = 1, 2, 3, \dots$, x real, put

$$f_n(x) = \frac{x}{1 + nx^2} \quad (5)$$

Then $\{f_n\}$ converges uniformly to a function f , and that the equation

$$f'(x) = \lim_{n \rightarrow \infty} f'_n(x) \quad (6)$$

is correct if $x \neq 0$, but false if $x = 0$.

Proof. To start, since $f_n(x)$ is a quotient of two polynomials and $1 + nx^2 > nx^2 \geq 0$, we have that $f_n(x)$ is differentiable for all x and all n . Thus

$$f'_n(x) = \frac{(1 + nx^2) - 2nx^2}{(1 + nx^2)^2} = \frac{1}{1 + nx^2} - \frac{2nx^2}{(1 + nx^2)^2} = \frac{1}{1 + nx^2} - \frac{2nx^2}{n^2x^4 + 2nx^2 + 1} \quad (7)$$

If $x = 0$, then $\frac{1}{1+nx^2} - \frac{2nx^2}{n^2x^4+2nx^2+1} = \frac{1}{1+n(0)^2} - \frac{2n(0)^2}{n^2(0)^4+2n(0)^2+1} = \frac{1}{1} = 1$ Thus if $x = 0$,

$$\lim_{n \rightarrow \infty} f'_n(x) = \lim_{n \rightarrow \infty} 1 = 1 \quad (8)$$

and if $x \neq 0$,

$$\lim_{n \rightarrow \infty} f'_n(x) = \lim_{n \rightarrow \infty} \frac{1}{1 + nx^2} - \frac{2nx^2}{n^2x^4 + 2nx^2 + 1} \quad (9)$$

We will now prove that the limit above goes to 0. First we prove that

$$\lim_{n \rightarrow \infty} \frac{1}{1 + nx^2} = 0. \quad (10)$$

For all $\epsilon > 0$, choose $N > \frac{1}{x^2\epsilon}$ (we do not need this limit to converge uniformly). Then $n > N \implies \left| \frac{1}{1+nx^2} \right| = \frac{1}{1+nx^2} < \frac{1}{nx^2} < \frac{1}{Nx^2} = \frac{1}{\frac{1}{x^2\epsilon}x^2} = \frac{1}{\epsilon} = \epsilon$. Now we prove that

$$\lim_{n \rightarrow \infty} \frac{2nx^2}{n^2x^4 + 2nx^2 + 1} = 0. \quad (11)$$

For all $\epsilon > 0$, choose $N > \frac{2}{x^2\epsilon}$ (we also do not need this limit to converge uniformly). Then $n > N \implies \left| \frac{2nx^2}{n^2x^4+2nx^2+1} \right| = \frac{2nx^2}{n^2x^4+2nx^2+1} < \frac{2nx^2}{n^2x^4} = \frac{2}{nx^2} < \frac{2}{Nx^2} = \frac{2}{\frac{2}{x^2\epsilon}x^2} = \frac{2}{\epsilon} = \epsilon$. Therefore for all x ,

$$\lim_{n \rightarrow \infty} \frac{1}{1 + nx^2} - \frac{2nx^2}{n^2x^4 + 2nx^2 + 1} = \lim_{n \rightarrow \infty} \frac{1}{1 + nx^2} - \lim_{n \rightarrow \infty} \frac{2nx^2}{n^2x^4 + 2nx^2 + 1} = 0 - 0 = 0. \quad (12)$$

Now we just need to find $f(x)$ and $f'(x)$. We will now show that $\{f_n\}$ uniformly converges to $f(x) = 0$. To do this we will first find the maximum of $f_n(x)$. Note that

$$f'_n(x) = \frac{(1 + nx^2) - 2nx^2}{(1 + nx^2)^2} = \frac{1 - nx^2}{(1 + nx^2)^2} \quad (13)$$

and so the critical points are where $1 - nx^2 = 0 \implies nx^2 = 1 \implies x^2 = \frac{1}{n} \implies x = \pm \frac{1}{\sqrt{n}}$. From the first derivative test, we have that $x = \frac{1}{\sqrt{n}}$ yields a maximum (since right below $x = \frac{1}{\sqrt{n}}$ the derivative is positive and right above $x = \frac{1}{\sqrt{n}}$ the derivative is negative). Thus the maximum for $f_n(x)$ is

$$\frac{\frac{1}{\sqrt{n}}}{1 + n(\frac{1}{\sqrt{n}})^2} = \frac{\frac{1}{\sqrt{n}}}{2} = \frac{1}{2\sqrt{n}} \quad (14)$$

and since $x < 0 \implies f_n(x) < 0$ we have that $\frac{1}{2\sqrt{n}}$ is indeed an absolute maximum for the function. Thus for any $\epsilon > 0$ take $N > \frac{1}{4\epsilon^2}$. Thus $|\frac{x}{1+nx^2}| = \frac{|x|}{1+nx^2} < \frac{1}{2\sqrt{n}} < \frac{1}{2\sqrt{\frac{1}{4\epsilon^2}}} = \frac{1}{2 \cdot \frac{1}{2\epsilon}} = \frac{1}{\epsilon} = \epsilon$. Thus $\{f_n\}$ uniformly converges to $f(x) = 0$. Thus $f'(x) = 0$. Therefore for $x \neq 0$, the equation

$$f'(x) = \lim_{n \rightarrow \infty} f'_n(x) \quad (15)$$

becomes $0 = 0$, which is correct, and if $x = 0$ this yields $0 = 1$, which is false. $\square$

Theorem 4. *If*

$$I(x) = \begin{cases} 0, & \text{if } x \leq 0 \\ 1, & \text{if } x > 0 \end{cases} \quad (16)$$

if $\{x_n\}$ is a sequence of distinct points of (a, b) , and if $\sum |c_n|$ converges, then the series

$$f(x) = \sum_{n=1}^{\infty} c_n I(x - x_n) \quad (a \leq x \leq b) \quad (17)$$

converges uniformly, and that f is continuous for every $x \neq x_n$.

Proof. Since $\sum_{n=1}^{\infty} c_n I(x - x_n) \leq \sum_{n=1}^{\infty} c_n$ and $\sum |c_n|$ converges, by the Weierstrass M-Test, we that $\sum_{n=1}^{\infty} c_n I(x - x_n)$ converges uniformly. Since for any $x \neq x_n$ we have that $c_n I(x - x_n)$ is continuous, it follows that $\sum_{n=1}^N c_n I(x - x_n)$ is continuous for any N , and since this series converges uniformly, we have that $f(x) = \sum_{n=1}^{\infty} c_n I(x - x_n)$ is continuous at every $x \neq x_n$. $\square$

Theorem 5. *Suppose $\mathcal{A}$ is an algebra such that $\mathcal{A}$ separates points and contains constants. Then for any two distinct points $x_1, x_2 \in E$ and any two complex constants c_1, c_2 there exists a function $f \in \mathcal{A}$ such that $f(x_1) = c_1$ and $f(x_2) = c_2$.*

Proof. Suppose $\mathcal{A}$ is an algebra such that $\mathcal{A}$ separates points. Thus for any two distinct points $x_1, x_2 \in E$ there exists a function $f \in \mathcal{A}$ such that $f(x_1) \neq f(x_2) \implies f(x_2) - f(x_1) \neq 0$. Thus we consider the function

$$u(x) = c_1 + [c_2 - c_1] \frac{f(x) - f(x_1)}{f(x_2) - f(x_1)}. \quad (18)$$

Then $u(x_1) = c_1 + [c_2 - c_1] \frac{f(x_1) - f(x_1)}{f(x_2) - f(x_1)} = c_1 + [c_2 - c_1](0) = c_1$ and $u(x_2) = c_1 + [c_2 - c_1] \frac{f(x_2) - f(x_1)}{f(x_2) - f(x_1)} = c_1 + [c_2 - c_1](1) = c_2$ as desired. Note that since $\mathcal{A}$ contains constants and is closed under addition and scalar multiplication we have that $u(x) \in \mathcal{A}$. $\square$

Theorem 6. *The set of all polynomials is an algebra.*

Proof. Suppose $\mathcal{A}$ is the set of polynomials and let $f, g \in \mathcal{A}$. Then $f = \sum_{i=0}^n a_i x^i$ and $g = \sum_{i=0}^m b_i x^i$ where $a_0, \dots, a_n, b_0, \dots, b_m$ are some constants and $m, n \in \mathbb{Z}^+ \cup \{0\}$. Assume that $m \leq n$ (if it not the case we can swap f and g) and extend $g = \sum_{i=0}^m b_i x^i = \sum_{i=0}^n b_i x^i$ where $b_{m+1} = \dots = b_n = 0$. Then $f + g = \sum_{i=0}^n (a_i + b_i) x^i$ which is a polynomial so $f + g \in \mathcal{A}$. Next $fg = \sum_{k=0}^{n+m} (\sum_{i+j=k} a_i b_j) x^k$ which is also a polynomial so $fg \in \mathcal{A}$. Finally, for any complex constant c , we have that $cf = c \sum_{i=0}^n a_i x^i = \sum_{i=0}^n c a_i x^i$ and so $cf \in \mathcal{A}$. Thus the set of polynomials is an algebra. $\square$

Theorem 7. *If we restrict the set of polynomials to only include polynomials of degree less than or equal to n where n is a finite integer, then the set ceases to be an algebra.*

Proof. Suppose $\mathcal{A}$ is the set of polynomials with the given restriction. Then there exists $f(x) = \sum_{i=0}^n a_i x^i$ and $g(x) = \sum_{i=0}^n b_i x^i$ with $a_n, b_n \neq 0$. Thus $fg = \sum_{k=0}^{2n} (\sum_{i+j=k} a_i b_j) x^k$ which is of degree $2n$ so $fg \notin \mathcal{A}$. Thus $\mathcal{A}$ is not an algebra. $\square$

Remark 1. *Theorem 7 highlights a key difference between a vector space and an algebra as the set of polynomials of degree n or less would be a vector space but as shown in Theorem 7 such a set is not an algebra.*

Theorem 8. *Suppose $\mathcal{A}$ is an algebra such that $\mathcal{A}$ separates points and contains constants. Then for n distinct points $x_1, x_2, \dots, x_n \in E$ and n complex constants $c_1, \dots, c_n$ there exists a function $f \in \mathcal{A}$ such that $f(x_i) = c_i$ for $1 \leq i \leq n$.*

Proof. We show this by induction. The base case is verified by theorem 5. For the inductive hypothesis, suppose there exists a function f such that for n distinct points $x_1, x_2, \dots, x_n \in E$ and n complex constants $c_1, \dots, c_n$, we have that $f(x_i) = c_i$ for $1 \leq i \leq n$. We want to show that for $n+1$ distinct points $x_1, x_2, \dots, x_n, x_{n+1} \in E$ and $n+1$ complex constants $c_1, \dots, c_n, c_{n+1}$, we have that $f(x_i) = c_i$ for $1 \leq i \leq n+1$. Since $\mathcal{A}$ separates points, for any x_i ($1 \leq i \leq n$) there exists a function $g_i(x)$ such that $g_i(x_i) \neq g_i(x_{n+1}) \implies g_i(x_{n+1}) - g_i(x_i) \neq 0$. Thus we consider the function

$$u(x) = f(x) + [c_{n+1} - f(x)] \prod_{i=1}^n \frac{g_i(x) - g_i(x_i)}{g_i(x_{n+1}) - g_i(x_i)}. \quad (19)$$

Then for $1 \leq j \leq n$, $u(x_j) = f(x_j) + [c_{n+1} - f(x_j)] \prod_{i=1}^n \frac{g_i(x_j) - g_i(x_i)}{g_i(x_{n+1}) - g_i(x_i)} = f(x_j) + [c_{n+1} - f(x_j)](0) = f(x_j) = c_j$ and $u(x_{n+1}) = f(x_{n+1}) + [c_{n+1} -$

$f(x_{n+1})] \prod_{i=1}^n \frac{g_i(x_{n+1}) - g_i(x_i)}{g_i(x_{n+1}) - g_i(x_i)} = f(x_{n+1}) + [c_{n+1} - f(x_{n+1})](1) = c_{n+1}$ as desired. Since $f \in \mathcal{A}$ and $\mathcal{A}$ is closed under addition, multiplication, and scalar multiplication, it follows that $u(x) \in \mathcal{A}$. $\square$

Remark 2. *Theorem 8 when applied to polynomials gives rise to Lagrange Interpolation where if you have two set of points $\{x_1, \dots, x_n\}$ and a continuous function f , you can approximate f using the polynomial*

$$P(x) = \sum_{k=1}^n f(x_k) \prod_{\substack{i=1 \\ i \neq k}}^n \frac{x - x_i}{x_k - x_i}. \quad (20)$$